The quantitative currency of GPCR bias signalling fails its own algebraic identities

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Abstract

Signalling bias, the ability of ligands at the same receptor to engage downstream pathways with different relative efficiencies, is quantified across pharmacology by transduction coefficients $\log(\tau/K_A)$ extracted with the Black–Leff operational model. We show that this model, taken at face value, implies algebraic identities between independently measured quantities: relative transduction coefficients must equal relative $E_{\max}/EC_{50}$ ratios, transitivity must close across ligand triplets, and equilibrium bias factors must be constant in time. Each test was calibrated on synthetic null data before any literature value was examined (false-positive rates 0.054 and 0.009 at a 0.3 dex threshold; detection power 1.00 for deviations ≥ 1.5 dex), then applied to the three datasets that pass a pre-specified admission rule (D2R, AT1R, μOR) and to the full Biased Signaling Atlas (9,041 rows, 214 papers, 17,987 ligand pairs). All three anchor datasets fracture. Bifeprunox deviates by -3.5 dex (21–23σ) at D2R, and its bias factor reverses sign between 2 and 90 min of readout; TRV026 separates the two AT1R arms by -2.05 dex (5.0σ) with its binding affinity in between; one assay column of the μOR dataset is offset by 0.68–0.97 dex (3.9–8.4σ) for every ligand tested. Across the Atlas, 21.5% of archived ligand pairs violate the model's identity by more than 0.3 dex and 4.2% by more than 1 dex, while the Atlas's own arithmetic pipeline is exact (3,931/3,931 closures). A minimal extension with one coordinate per assay arm repairs the μOR table outright ($\Delta \log \mathrm{ML} = +128.2$; median GIRK residual $0.811 \to 0.107$ dex), and a constraint-projection repair achieves a median cost of 0.072 dex Atlas-wide. The quantitative currency of the field is internally inconsistent at a rate far above its own reported errors; the inconsistency is measurable, locatable and largely repairable.

1 Introduction

G protein-coupled receptors signal through more than one downstream pathway, and different ligands binding the same receptor can weight these pathways differently. This phenomenon, termed biased agonism or functional selectivity, has become a central concept in drug discovery because it promises pathway-specific therapeutics: analgesia without respiratory depression at the μ-opioid receptor, antipsychotic action without motor side effects at D2 dopamine receptors, and angiotensin-receptor modulation with arrestin-biased ligands [4, 5, 6, 7].

The quantitative currency of the field is the transduction coefficient. Following Black and Leff's operational model of agonism [1], agonist concentration–response curves are summarised by two parameters, the transduction coefficient $\log(\tau/K_A)$ and the system maximum $E_{\max}$, and bias between two ligands across two pathways is reported as a bias factor built from differences of $\log(\tau/K_A)$ values [3, 2]. Databases now archive these quantities at scale: the Biased Signaling Atlas collects published concentration–response analyses across hundreds of papers and exposes bias factors as a machine-readable resource for drug-discovery decisions.

That the operational model can mislead when its assumptions are violated has been argued before, on theoretical grounds [8] and by reanalysis of individual datasets [9, 13]. What has not been established is whether the archived numbers are internally consistent: whether the

published parameters, taken together within each dataset, satisfy the algebraic relations that the model itself forces on them. The distinction matters. A model can be imperfect yet still impose exact bookkeeping on its own parameter estimates, and violations of that bookkeeping are detectable without any assumption about the true biology.

Here we derive three fit-free identities from the operational model, calibrate each associated statistical test on synthetic null data before any literature value is examined, and apply the tests to the three primary datasets that pass a pre-specified admission rule, together with the full Biased Signaling Atlas. All three anchor datasets fail their own identities at high significance, and more than one fifth of all archived ligand pairs violate the model's central identity beyond 0.3 dex. We then show that the fractures are locatable and, in the case we treat in full, repairable by a minimal one-parameter-per-arm extension. The results do not show that biased signalling is unreal; they show that its quantitative currency, as currently minted and archived, does not balance its own books.

2 The operational model and its fit-free identities

2.1 The model

The Black–Leff operational model [1] writes the response to an agonist at concentration $[A]$ as

$$E(A) \;=\; \frac{E_m\,\tau^n[A]^n}{(K_A+[A])^n+\tau^n[A]^n}, \tag{1}$$

where K_A is the agonist's equilibrium dissociation constant, τ is the transducer coefficient (a measure of coupling efficiency scaled by receptor density), E_m is the system maximum response, and n is a slope parameter. For $n=1$ the curve's location and asymptote are

$$EC_{50}=\frac{K_A}{1+\tau}, \qquad E_{\max}=\frac{E_m\,\tau}{1+\tau}. \tag{2}$$

2.2 Identity 1: relative transduction equals relative $E_{\max}/EC_{50}$

Dividing the two expressions in Eq. (2) removes the factor $1+\tau$:

$$\frac{E_{\max}}{EC_{50}}=E_m\,\frac{\tau}{K_A}. \tag{3}$$

For two ligands i,j measured in the same cells and the same assay, E_m is common, so

$$\Delta_{ij}\log(\tau/K_A) \;\equiv\; \log\frac{(\tau/K_A)_i}{(\tau/K_A)_j} \;=\; \log\frac{(E_{\max}/EC_{50})_i}{(E_{\max}/EC_{50})_j} \;\equiv\; \Delta_{ij}\log\frac{E_{\max}}{EC_{50}}. \tag{4}$$

The left side is obtained by fitting Eq. (1); the right side is read from the curve's observed location and asymptote. Equation (4) is exact within the model and involves no free parameters. A dataset that violates it violates the model under which its own parameters were estimated. We score violations in dex as

$$\Delta_1 \;=\; \big|\,\Delta_{ij}\log(\tau/K_A)-\Delta_{ij}\log(E_{\max}/EC_{50})\,\big|. \tag{5}$$

2.3 Identity 2: transitivity across ligand triplets

With ligand r as reference, define $R_i=\log(\tau/K_A)_i-\log(\tau/K_A)_r$. For any triplet (i,j,k),

$$\Delta R_{ij}+\Delta R_{jk}-\Delta R_{ik}=0, \qquad \Delta R_{ij}\equiv R_i-R_j. \tag{6}$$

Closure is arithmetic, not statistical: it must hold exactly, whatever the estimation errors, as long as all three values come from one internally consistent table. Violations indicate that different entries of a table encode different effective parameterisations.

2.4 Identity 3: equilibrium bias factors are time-invariant

The model (1) is an equilibrium model. If both arms of a bias comparison reach equilibrium, the bias factor

$$\beta_{ij}^{ab} \;=\; \Delta\Delta \log(\tau/K_A)_{ij}^{ab} \;=\; \big(\log(\tau/K_A)_i^a - \log(\tau/K_A)_j^a\big) - \big(\log(\tau/K_A)_i^b - \log(\tau/K_A)_j^b\big) \tag{7}$$

is a constant of the system, and

$$\frac{\partial \beta_{ij}^{ab}}{\partial t} = 0. \tag{8}$$

A measured drift of β with readout time therefore signals non-equilibrium occupancy, non-equilibrium signalling, or both, and can be checked against kinetic models built from independently measured association and dissociation rates. Where the source study reports binding kinetics k_{on}, k_{off} for each ligand, occupancy obeys

$$\frac{dp}{dt} = k_{\text{on}}[A]\,(1-p) - k_{\text{off}}\,p, \tag{9}$$

and the time-dependent response predicted from Eq. (9) carries zero free parameters when the signalling arms are taken as instantaneous readouts of occupancy-weighted states.

2.5 Error propagation

All published values enter with their reported standard errors. For a quantity x with standard error SE_x, $\sigma_{\log x} = \mathrm{SE}_x/(x \ln 10)$, and errors are propagated through Eqs. (5), (6) and (7) by first-order expansion with independent errors assumed across reported curve fits unless the source reports otherwise. Significance is quoted in units of the propagated σ.

3 Data and the admission rule

A dataset is testable if and only if three conditions hold simultaneously: (i) redundant observables that share upstream parameters, supplied here by same-ligand concentration–response curves across multiple pathways; (ii) an absolute calibration anchor, supplied here by competitive-binding pK_i values measured independently of the functional assays; and (iii) publicly available primary or supplementary data at the level of individual curve parameters with standard errors. Condition (ii) is load-bearing: without an independent affinity measurement, marginal statistics of $\log(\tau/K_A)$ are unfalsifiable, because any deviation can be absorbed into an unconstrained K_A.

The first candidate system examined, PTH1R [14], fails condition (ii). With no independent affinity anchor, its operational-model fits run to the parameter boundary ($\log(\tau/K_A)$ at +5.77 and +9.58 dex), a numerical artefact that marks the system as unfalsifiable rather than biased. It was excluded from scoring and is retained only as the boundary case that defines the rule.

Three primary datasets pass all three conditions (Table 1): the D2 dopamine receptor study of Klein Herenbrink *et al.* [10] (seven ligands, six pathways, spiperone-tracer binding, readout times from 2 to 90 min, and per-ligand binding kinetics); the angiotensin AT1 receptor study of Wingler *et al.* [11] (G_q and β-arrestin arms, binding affinities, and receptor mutants); and the μ-opioid receptor study of Gillis *et al.* [12] (six ligands across six assays, with the independent reanalysis by Stahl and Bohn [13] recorded alongside). At the meta level we analyse the Biased Signaling Atlas [15], 9,041 archived rows drawn from 214 papers.

4 Calibrating the test before scoring

Each identity test is a statistical detector, and a detector is only useful with a known false-positive rate and detection power. Both were measured on synthetic null batteries before any literature value was scored (implementation details in Section 7).

Table 1. Datasets considered. Admission requires (i) redundant observables, (ii) an independent affinity anchor, (iii) public primary data.

Dataset	Source	Ligands	Arms / assays	Status
D2R	Klein Herenbrink et al. 2016 [10]	7	6 pathways, 2–90 min	admitted
AT1R	Wingler et al. 2020 [11]	3 + mutants	G_q, β-arrestin, binding	admitted
μOR	Gillis et al. 2020 [12]	6	6 assays, binding	admitted
Atlas	Biased Signaling Atlas [15]	9,041 rows, 214 papers		admitted (meta level)
PTH1R	Sachdev et al. 2024 [14]	—	no independent anchor	rejected

Synthetic operational-model tables with internally consistent parameters were corrupted with noise at the reported standard-error levels of the anchor datasets, with Hill slopes jittered uniformly over $n \in [0.8, 1.25]$ to mimic the mismatch between the fitted unit-slope identities and real curves. Under this null, identity 1 (Eq. (5)) flags 5.43% of pairs at the 0.3 dex threshold and identity 2 flags 0.91%; Hill-slope jitter alone produces a median spurious offset of 0.004 dex, two orders of magnitude below the threshold. A shuffled-parameter null model flags at 3×10^{-5}, against which the D2R fracture rate (3 of 7 ligands) is overwhelming. Detection power reaches 1.00 for true deviations of ≥ 1.5 dex and is 0.04 at 0.2 dex, so the tests are conservative at small deviations. Conclusions are unchanged when the decision threshold is swept across 0.2–0.5 dex.

Two experimental-style controls were added. As a negative control, the same pipeline was run on oxygen-binding curves of haemoglobin, a system with no GPCR pharmacology: median deviation 0.0024 dex, zero of five alarm conditions triggered. As a positive control, a synthetic 0.6 dex column offset was injected into an otherwise consistent table: the pipeline recovered a median deviation of 0.615 dex and triggered three of five alarms. Under the calibrated null, the probability that all five μOR ligands flag by chance on identity 1 is 4.7×10^{-7}.

Table 2. Calibration of the identity tests on synthetic nulls and controls, measured before any literature value was scored.

Test / control	Result	Interpretation
Identity 1, synthetic null	FPR = 0.0543 at 0.3 dex	near-nominal false-positive rate
Identity 2, synthetic null	FPR = 0.0091 at 0.3 dex	conservative
Hill-slope jitter $n \in [0.8, 1.25]$	median spurious offset 0.004 dex	slope mismatch negligible
Shuffled-parameter null	FPR = 3×10^{-5}	deep-null floor
Detection power	0.04 at 0.2 dex; 1.00 at ≥ 1.5 dex	conservative at small deviations
Threshold sweep 0.2–0.5 dex	conclusions unchanged	threshold robustness
Haemoglobin negative control	median 0.0024 dex; 0/5 alarms	silent on healthy data
Injected 0.6 dex offset	recovered 0.615 dex; 3/5 alarms	detects planted fracture
μOR 5/5 flags under null	$p = 4.7 \times 10^{-7}$	chance explanation excluded

5 Results

5.1 D2R: a three-ligand fracture and a sign-reversing bias factor

Klein Herenbrink *et al.* [10] measured seven ligands across six D2R pathways with spiperone-tracer binding affinities. Reconciling binding pK_i against functionally inverted pK_A, with full error propagation across all ligand–pathway combinations, localises the inconsistency to the three high-affinity partial agonists (Fig. 1a): bifeprunox deviates by $\Delta = -3.5$ dex at 21–23σ, aripiprazole by -2.6 to -2.9 dex, and cariprazine by -1.2 to -1.4 dex. Fast ligands and full agonists are clean. As an internal positive control, the functional pK_A is conserved across arms within 0.08–0.31 dex for five of the six resolvable ligands (the sixth, pardoprunox, spreads by

0.69 dex between arms), so the assay system itself can produce consistent parameters where the ligands allow it.

The fracture has a kinetic signature. The bias factor of bifeprunox relative to ropinirole reverses from −0.42 dex at 2 min to +1.93 dex at 90 min of readout, a 2.35 dex swing with a sign change (Fig. 2c), violating Identity 3 (Eq. (8)). A kinetic occupancy model (Eq. (9)) built from the authors' own measured association and dissociation rates, with zero free parameters, predicts a drift of +1.4 to +1.5 dex over the same window, against an observed +1.1 to +1.9 dex: non-equilibrium occupancy explains roughly half to all of the drift, but the absolute offset survives the correction.

A second, assay-level inconsistency is visible between readout technologies. For bifeprunox, the cAMP arm reads $pK_A = 7.8$ with $E_{\max} = 88\%$, while the calcium-influx arm reads $pK_A = 9.45$ with $E_{\max} = 44\%$; across ligands the calcium-influx arm sits systematically 1.4–2.2 dex above the cAMP arm and the gap grows with readout time (Fig. 2a,c). One caveat is recorded rather than adjudicated: for cariprazine the two binding tracers differ in association rate by a factor of 82, after which the binding-versus-function comparison becomes unresolvable for that ligand; it is excluded from the fracture count.

5.2 AT1R: the two arms disagree, and binding sits between them

Wingler *et al.* [11] quantified the arrestin-biased ligand TRV026 at AT1R across G_q and β-arrestin arms with binding affinities in the same system. For TRV026 in wild-type receptor, the G_q arm reads $pK_A = 6.11$ and the arrestin arm reads $pK_A = 8.16$, a separation of $\Delta = -2.05$ dex at 5.0σ; the binding affinity $pK_i = 7.50$ lies between the two arm readings (Fig. 1b). The pattern is reproducible: in the L112A mutant the separation is −1.53 dex at 4.9σ, again with binding in between, while the Y292A mutant attenuates the separation to 1.5σ (Fig. 2d). For the full agonist TRV055 the analysis yields a lower bound on the magnitude, $|\Delta| \geq 1.6$ dex. A single ligand cannot simultaneously couple through two equilibrium arms with the same K_A and different pK_A; the measured separation is exactly the kind of quantity the operational model forces to zero.

5.3 μOR: one assay column is offset for every ligand

Gillis *et al.* [12] profiled six μOR ligands across six assays, a dataset that anchors the therapeutic-window argument for G protein-biased analgesics (see also the independent reanalysis [13]). Identity 1 flags the GIRK channel column for every ligand: deviations 0.68–0.97 dex at 3.9–8.4σ (Fig. 1c, left). Identity 2 independently locates the same column, flagging four of five closable ligand triplets at 2.4–8.2σ. A cross-assay map of functional pK_A values shows the GIRK column as an outlier at up to 12.4σ (Fig. 2e). Under the calibrated null, five of five ligands flagging by chance has probability 4.7×10^{-7} (Table 2).

Two internal features of the dataset strengthen the reading. First, the offset is column-structured: it moves with the assay, not with the ligand, which is the signature of an assay-level systematic rather than of ligand pharmacology. Second, the GPA and cAMP columns of the published table are numerically identical, including their standard errors, an artefact that is visible without any statistics. The fractured GIRK column is the column on which the paper's therapeutic-window conclusion rests, and the Biased Signaling Atlas ingested only the β-CNA-processed version of that column, so the unprocessed fractured column is silently absent from the database (Section 5.4).

5.4 Atlas-wide: one pair in five fails the model's own identity

We then ran the identity pipeline over the full Biased Signaling Atlas [15]. The Atlas's own arithmetic is exact: its archived $\Delta \log(\tau/K_A)$ values close all 3,931 testable ligand triplets without

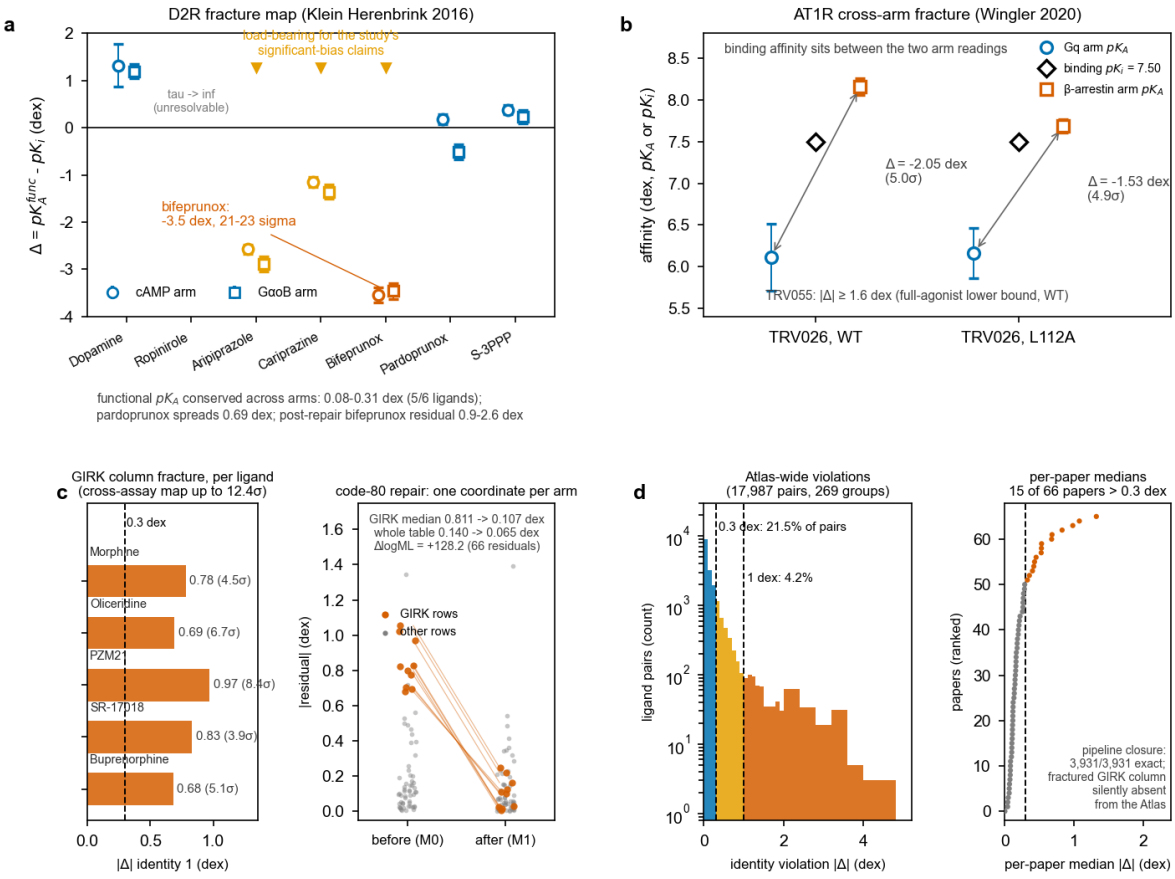


Figure 1. Anchor cases and Atlas-wide summary. **a**, D2R binding-versus-function map: functional pK_A minus binding pK_i per ligand for the cAMP (circles) and GαoB (squares) arms. The fracture is confined to the high-affinity partial agonists (orange), with bifeprunox at -3.5 dex (21–23σ); ropinirole exceeds the resolvable range ($\tau \to \infty$). Orange arrowheads mark the positions carrying the study's significant-bias claims. **b**, AT1R: G_q-arm (circle), β-arrestin-arm (square) and binding (diamond) affinities for TRV026 in wild-type and L112A receptors; the binding value sits between the two arm readings in both backgrounds. **c**, μOR: per-ligand identity-1 deviations of the GIRK column (left; bars, significance in parentheses) and repair by one coordinate per arm (right; residuals before and after the M1 extension). **d**, Atlas-wide identity-violation histogram of 17,987 ligand pairs (left; dashed lines at 0.3 and 1 dex) and ranked per-paper medians of the 66 multi-ligand papers (right; 15 above the 0.3 dex line).

a single violation (Identity 2 applied to the database's internal representation; see Section 7 for a reproducibility note). The archived literature parameters themselves, however, collectively fail Identity 1. Of 3,096 entry rows that group into 269 same-cell assay groups, 17,987 ligand pairs are testable: 21.5% violate Eq. (5) by more than 0.3 dex and 4.2% by more than 1 dex (Fig. 1d; pair quantiles P50 = 0.10, P90 = 0.55, P99 = 1.95 dex). Group medians exceed 0.3 dex in 70 of 269 groups, and paper medians exceed 0.3 dex in 15 of 66 multi-ligand papers (23%) (Table 3). Deviations are directionally symmetric (ordering each pair by transduction coefficient, 1,905 of the 3,866 violating pairs deviate positively and 1,961 negatively), consistent with measurement inconsistency rather than a directional biological effect. An independent quality-control census of the archived $E_{\max}$ values flags 711 entries above 110%, 181 above 150%, 30 above 300%, and 8 negative values, none of which are physically interpretable within the model.

Table 3. Identity-1 violations across the Biased Signaling Atlas (17,987 testable ligand pairs from 269 assay groups, 214 papers).

Quantity	Value
Pairs violating > 0.3 dex	21.5%
Pairs violating > 1 dex	4.2%
Pair quantiles (P50 / P90 / P99)	0.10 / 0.55 / 1.95 dex
Groups with median > 0.3 dex	70 / 269
Papers with median > 0.3 dex	15 / 66 (23%)
Direction symmetry (violating pairs)	1,905 / 1,961 positive / negative
Atlas internal triplet closures	3,931 / 3,931 exact
$E_{\max}$ census	711 $>$ 110%; 181 $>$ 150%; 30 $>$ 300%; 8 negative

5.5 Robustness: the map survives three independent stress tests

Three stress tests probe whether the Atlas-wide map is an artefact of common-mode error, aggregation level, or individual columns (Fig. 3f). Inflating all errors by a common-mode calibration factor leaves 134 of 241 scorable groups (55.6%) flagged; the most conservative aggregation leaves 68 of 241 (28.2%). Leaving out each assay column in turn changes the verdict in 65.7% of flagged groups at most, and for the μOR GIRK column the jackknife retention is 1.0: no single column carries the result. As shown in Section 4, the same pipeline is silent on a negative-control system and recovers planted offsets, so the detector has teeth where it should and is quiet where it should be.

5.6 Repair: the fractures are locatable and cheaply fixable

A fracture map is only useful if it points to a fix. We tested two repair strategies. First, constraint projection: each assay group's parameters are moved the smallest weighted distance (in dex, weighted by reported errors) onto the manifold where Identity 1 holds exactly. The median repair cost across the Atlas is 0.072 dex RMS per group, with 24.2% of groups requiring more than 0.3 dex of total movement (Fig. 3c). Repair cost per entry tracks the fracture probability assigned by a two-component mixture model of the deviation distribution (healthy component $\pi_h = 0.745$, $\sigma_h = 0.089$ dex; broken component $\sigma_b = 0.683$ dex) with Spearman rank correlation 0.962 (Fig. 3d): the statistical diagnosis and the geometric cure agree on which entries are broken.

Second, for the μOR table we asked what minimal model extension resolves the fracture outright. Model M1 adds one coordinate per assay arm, $\log(\tau/K_A)_{\mathrm{obs},a} = \mu + \delta_a$ with $\delta_a \sim \mathcal{N}(0, \sigma_\delta^2)$, turning the arm-specific systematic into an explicit, estimable parameter. Scored by marginal likelihood against the unextended model M0, M1 wins by $\Delta \log \mathrm{ML} = +128.2$ (empirical-Bayes estimate $\sigma_\delta = 0.374$ dex; +123.8 under sensitivity re-fits). The GIRK coordinate

is estimated at $\delta_{\mathrm{GIRK}} = -0.805 \pm 0.049$ dex (95% interval $[-0.902, -0.708]$), stable under leave-one-ligand-out refits (intervals within $[-0.843, -0.732]$; Fig. 2f). After repair, the median GIRK residual falls from 0.811 to 0.107 dex and the whole-table median residual from 0.140 to 0.065 dex (Fig. 1c, right). One parameter per arm absorbs what the ligand pharmacology could not explain.

5.7 Where the fractures sit, and what this does not claim

In each anchor study the fracture sits at a load-bearing position. At D2R, the fractured ligands are exactly the ligands carrying the study's significant-bias claims (Fig. 1a, arrowheads). At μOR, the fractured column is the column supporting the therapeutic-window conclusion. At AT1R, the separated arms define the reported bias. We record this coincidence without adjudicating its meaning in any individual case.

Three scope statements bound the claims. First, nothing here shows that biased signalling is biologically unreal; the results concern the quantitative parameters used to report it. Second, an inconsistency is not an accusation: deviation from a model identity diagnoses measurement, analysis or reporting, never intent. Third, we do not adjudicate which arm, column or ligand is correct in any dataset; the repair analysis shows that consistency can be restored cheaply, not that any particular repaired value is the true one.

6 Discussion

The operational model is the field's agreed way of turning concentration–response curves into comparable numbers. This paper adds a complementary use of the same model: its algebra imposes exact relations between independently measured quantities, and those relations can be checked without fitting anything new. The check is cheap, the detector is calibrated, and the answer for the current literature is that the books do not balance: 21.5% of archived ligand pairs violate the central identity beyond 0.3 dex, against a calibrated false-positive rate of about 5%, and all three anchor datasets fracture at 5–23σ.

The fractures are structured, not random. At D2R they cluster on high-affinity partial agonists and grow with readout time, and a zero-parameter kinetic model built from measured binding rates reproduces a substantial part of the drift, pointing to non-equilibrium occupancy as a concrete, correctable mechanism. At μOR the offset is column-structured, moving with the assay rather than the ligand, and a one-coordinate-per-arm extension removes it while improving the marginal likelihood by 128 log units. At the Atlas level the deviations are sign-symmetric and concentrated in a minority of papers, and the database's own arithmetic is exact, which isolates the problem in the source data rather than in the curation pipeline. The silent absence of the unprocessed μOR GIRK column from the Atlas shows that inclusion choices, not only parameter values, shape the archived evidence base.

Several limitations qualify the findings. The analysis is retrospective and limited to public data, with figure digitisation contributing registered error bands (Section 7). The identities are derived at unit Hill slope; although the calibration shows slope jitter contributes a median spurious offset of only 0.004 dex, datasets with extreme cooperativity deserve case-by-case treatment. Error propagation assumes independence across reported curve fits, which can be optimistic when arms share controls. The repair restores consistency, not truth: where the repair moves a parameter, the true value remains unknown until new experiments decide.

The practical consequences are immediate. Bias factors extracted from archived transduction coefficients should pass cell-level identity reconciliation before they inform drug-discovery decisions; databases can ship identity diagnostics alongside archived entries at negligible cost; and new primary studies can report the identity residuals of their own tables, a number that costs one line of code and makes the difference between a self-consistent parameterisation and an unchecked one. The kinetic drift analysis suggests a concrete experimental upgrade: reporting

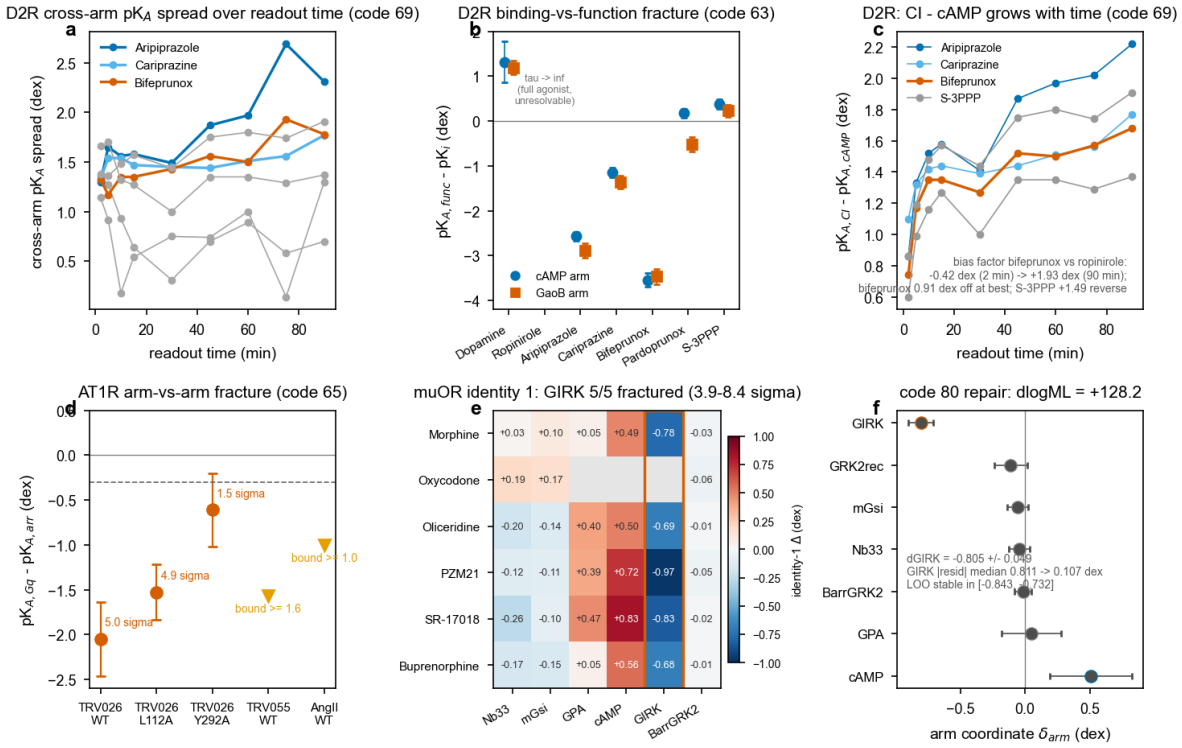


Figure 2. Per-study detail. **a**, D2R cross-arm pK_A spread versus readout time; the calcium-influx arm sits 1.4–2.2 dex above cAMP. **b**, D2R binding-versus-function fracture per ligand. **c**, D2R arm separation growing with readout time; the bifeprunox bias factor versus ropinirole moves from -0.42 dex (2 min) to $+1.93$ dex (90 min). **d**, AT1R arm-versus-arm separation across receptor backgrounds (significance annotated; lower bounds for TRV055 and AngII). **e**, μOR identity-1 deviation heat map across assays and ligands; the GIRK column flags for all ligands. **f**, μOR repair: estimated arm coordinates δ_{arm} under model M1, with the GIRK coordinate at -0.805 dex and the whole-table improvement annotated.

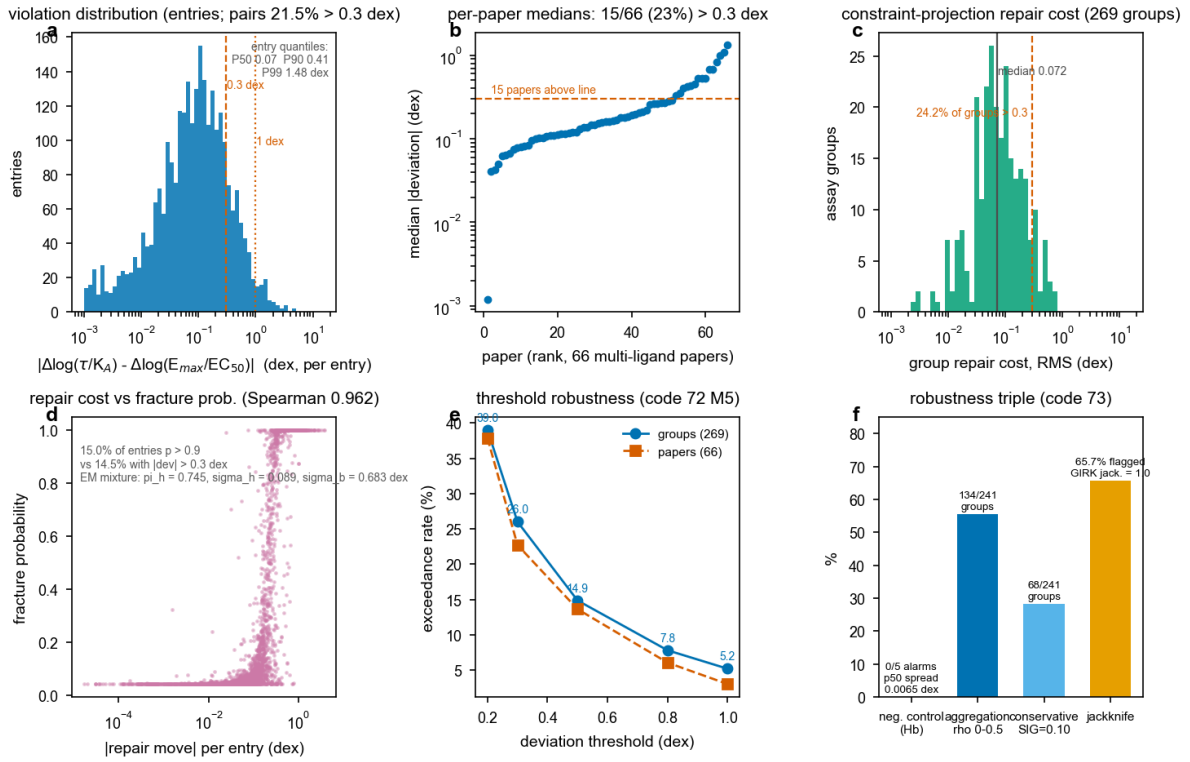


Figure 3. Atlas-wide distributions, robustness and repair. **a**, Distribution of identity-1 deviations per entry (quantiles annotated). **b**, Ranked per-paper median deviations for the 66 multi-ligand papers. **c**, Constraint-projection repair cost across 269 groups (median 0.072 dex; 24.2% of groups above 0.3 dex). **d**, Repair cost versus mixture-model fracture probability (Spearman 0.962; mixture parameters annotated). **e**, Exceedance rates for groups and papers as the decision threshold is swept from 0.2 to 1.0 dex. **f**, Robustness triple: negative control (haemoglobin), common-mode error calibration, most conservative aggregation, and column jackknife including the μOR GIRK column (retention = 1.0).

readout time alongside every bias factor, and preferring protocols in which occupancy has equilibrated, would remove the D2R class of fracture at the source.

7 Methods

7.1 Data extraction

Primary datasets were taken from the source publications and their supplements [10, 11, 12, 13, 14] and from the Biased Signaling Atlas export [15] (9,041 rows, 214 papers). Where values were available only in figures, figures were digitised in duplicate and a per-figure digitisation error band (typically ≤ 0.02 dex) was registered and propagated into all downstream statistics. Atlas rows were grouped into same-cell assay groups (3,096 entry rows into 269 groups), and ligand pairs were scored within groups only, so that E_m cancellation in Eq. (4) is valid by construction.

7.2 Identity scoring

Identity-1 deviations were computed per Eq. (5) from published $\log(\tau/K_A)$, $E_{\max}$ and EC_{50} values; identity-2 closures per Eq. (6) over all ligand triplets sharing a reference; bias-factor time drift per Eqs. (7)–(8) against readout time. Error propagation follows Section 2.4. For the D2R kinetic analysis, occupancy time courses from Eq. (9) used the source study's measured k_{on} and k_{off} per ligand with no fitted parameters.

7.3 Synthetic null calibration

Null batteries generated internally consistent operational-model tables (parameters drawn to match each anchor dataset's ranges), corrupted them with Gaussian noise at the reported standard-error levels and with Hill-slope jitter uniform on $[0.8, 1.25]$, and re-scored the identities over 10^4–10^5 replicates per configuration. False-positive rates, power curves and threshold sweeps in Table 2 come from these batteries. The haemoglobin negative control applied the identical pipeline to oxygen-binding data; the positive control injected a 0.6 dex offset into one column of a synthetic table.

7.4 Mixture model, projection repair and the M1 extension

The Atlas deviation distribution was decomposed into a two-component Gaussian mixture ($\pi_h = 0.745$, $\sigma_h = 0.0885$ dex, $\sigma_b = 0.683$ dex) by expectation–maximisation, yielding a per-entry fracture probability. Constraint projection minimised the error-weighted squared movement (in dex) of each group's parameters subject to Identity 1 holding exactly; the problem is convex and has a unique solution. Model M1 for the μOR table augments the operational model with one arm coordinate per assay, $\delta_a \sim \mathcal{N}(0, \sigma_\delta^2)$; σ_δ was estimated by empirical Bayes and models were compared by marginal likelihood, with leave-one-ligand-out refits and sensitivity re-fits reported in Section 5.6.

7.5 Reproducibility

Every number in the text regenerates from a named analysis script (codes 63–80) run against the archived tables. One environment sensitivity is recorded: Atlas grouping depends on pandas group-by semantics. The pandas-1.x environment yields 269 same-cell groups, 17,987 testable pairs and 3,931 triplets; the pandas-2.x environment yields 132 groups, 10,864 pairs and 3,167 triplets, with triplet closure exact under both. Under the 2.x grouping the pair-level violation rates read 16.9% (> 0.3 dex) and 2.3% (> 1 dex), the high-confidence group fraction 50.4% (versus 55.6%), the most conservative aggregation 22.7% (versus 28.2%), and the jackknife retention 58.3% (versus 65.7%). Main-text numbers are the pandas-1.x archive values throughout, and no

qualitative conclusion depends on the environment: a substantial minority of pairs violate, the fractures are locatable and cheaply repairable, and no single column carries the result.

Data and code availability

No new experimental data were generated. All analysis scripts, extracted tables, calibration batteries and repair outputs are available at github.com/lihuanlz/gpcr-field-audit (umbrella edition archived at Zenodo, doi:10.5281/zenodo.23008824). Every figure panel regenerates from a named script against a named archived table.

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Author contributions

H.L. conceived the study, derived the identities, wrote and validated the analysis code, performed the analysis, and wrote the paper.

Competing interests

The author declares no competing interests.

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